

Solar Microgrid Technical Guide

Project Overview

The pilot microgrid supplies a remote engineering laboratory. The system combines a 24 kW photovoltaic array, a 40 kWh lithium-iron-phosphate battery bank, and a 15 kW bidirectional inverter. The design target is to keep essential laboratory loads powered for at least 8 hours during a grid outage.

Operating Limits

The battery should normally operate between 20% and 90% state of charge (SOC). If SOC falls below 15%, non-essential loads are disconnected automatically. The inverter shuts down if its internal temperature reaches 85 C. Normal inverter efficiency is approximately 96%.

Control Strategy

During daylight, solar power serves the load first. Surplus solar power charges the battery. If the battery reaches 90% SOC, additional surplus may be exported to the grid. During an outage, grid export is disabled and the system operates in island mode. Essential loads are supplied from solar and battery power.

Maintenance Schedule

Task	Interval	Responsible Team
Visual inspection of PV modules	Monthly	Facilities
Battery health report	Every 3 months	Electrical Engineering
Inverter firmware review	Every 6 months	Controls Team
Full protection-system test	Annually	Electrical Engineering

Fault Codes

F12 indicates battery undervoltage. F27 indicates inverter overtemperature. F31 indicates loss of communication with the battery-management system. For F27, the operator should reduce load, verify ventilation, and allow the inverter to cool before restart.

Data Logging

The controller records DC bus voltage, battery SOC, inverter temperature, solar power, and load power every 10 seconds. Logs are retained locally for 30 days. The guide does not specify a cloud-storage provider or an external database service.